



Preliminary Assessment of Gold Potential: Staked Claims Area, Barstow Region, San Bernardino County, California

This preliminary high-level assessment evaluates the gold potential of the staked claims located southwest of Barstow, California, specifically west of Hodge and north of Helendale, in proximity to Iron Mountain. The presence of both lode and placer gold claims indicates potential for a dual exploration approach, which is characteristic of historic mining districts in the Mojave Desert. The underlying geology, comprising Mesozoic plutonic and metavolcanic rocks, presents a favorable yet complex environment for gold mineralization, as documented in publications by the California Geological Survey (CGS) regarding regional mineral resources.

The geologic mapping of the claims area identifies three primary lithologic units relevant to gold exploration. The predominant unit within the claim grid is designated "gb," consisting of Mesozoic plutonic rocks, primarily gabbro and dark dioritic compositions. Although felsic intrusions such as granite (associated with the adjacent GrMz unit) are more commonly linked to classic gold-bearing quartz veins, mafic rocks like gabbro and diorite can also host significant gold mineralization. This occurs due to their brittle response to tectonic stresses and chemical reactivity with hydrothermal fluids, facilitating the development of gold-bearing veins along structural weaknesses, such as shear zones or fractures within or proximal to plutonic contacts. In certain districts, gabbroic bodies may contribute thermal drive or fluid pathways for gold deposition, while iron-rich phases can serve as reductants, promoting gold precipitation from solution.

The adjacent GrMz unit comprises Mesozoic granite, quartz monzonite, and granodiorite—prototypical host rocks for lode gold deposits across the western United States, including historic occurrences in the Mojave Desert region, as detailed in U.S. Geological Survey (USGS) reports on gold in San Bernardino County.

The juxtaposition of mafic (gb) and felsic (GrMz) Mesozoic intrusions within the claim block thus offers multiple geologic models for lode gold emplacement.

Contacts between the gb and GrMz units are of particular significance, as these boundaries often exhibit enhanced permeability, channeling hydrothermal fluid migration. Structural controls, including faults and fracture systems, are expected to dominate gold deposition, concentrating mineralization within discrete zones rather than disseminated throughout massive rock volumes.

The Mv unit, consisting of pre-Cenozoic metavolcanic rocks (including latite, dacite, tuff, and greenstone), lies somewhat peripheral but represents a regionally prospective lithology for various mineralization styles, including gold in altered greenstone sequences.

Any lode-hosted gold identified within the gb or GrMz units would serve as the primary source for associated placer deposits in the area. Placer gold derives from mechanical erosion and gravitational concentration of lode sources in alluvial systems, a relationship supported by historical mining records in the Barstow-Iron Mountain vicinity.

Conclusions and Recommended Next Steps

The claims area exhibits moderate gold potential, attributable to its position within the broader Mojave Desert mineral province and the presence of Mesozoic plutonic assemblages known to host lode deposits in analogous settings. The combination of mafic and felsic intrusions supports multiple targeting opportunities.

A dual-pronged exploration strategy is recommended: (1) targeting structurally controlled quartz veins and potential disseminated mineralization within the intrusive rocks for lode gold; and (2) evaluating alluvial drainages and gravels for placer concentrations.

The immediate next phase should comprise a targeted field reconnaissance program, including detailed structural mapping and prospecting focused on gb-GrMz contacts, shear zones, and brittle



fracture networks. Systematic rock chip sampling of altered or veined outcrops is advised, with sample preparation, digestion, and analytical methods to be determined in consultation with qualified laboratory personnel prior to submission.

For placer evaluation, a preliminary program of systematic stream sediment sampling or dry concentration techniques in lower-elevation alluvial-covered portions of the claims is warranted, particularly in areas of thicker cover. Positive results would justify advancement to mechanical trenching or shallow drilling to delineate potential sources.

References

1. California Department of Conservation, California Geological Survey (CGS). Various publications on Mojave Desert geology and mineral resources.
2. U.S. Geological Survey (USGS). Various reports on gold and mineral occurrences in San Bernardino County, California.
3. Historical mining records for the Barstow-Iron Mountain region, San Bernardino County, California (including past operator reports and state/federal mineral databases).